

Question Paper

Exam Date & Time: 18-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Ethereum Smart Contract Using Solidity [IT_4421]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) In a blockchain network, each block contains the hash of the previous block along with the transaction data. The hash value is recalculated each time a new block is added. Apply the concept of cryptographic hash functions to explain how they are used to maintain the integrity of the blockchain. What would happen if someone tried to modify the data in an existing block? (5)
- A)

Course Outcome: "CO1" / Cognitive Level: "Application"

- B) A startup is developing a secure messaging application that allows users to exchange confidential messages over the internet. The development team decides to use asymmetric cryptography to protect message confidentiality and verify sender authenticity. Construct a security mechanism using the principles of asymmetric cryptography that ensures both confidentiality and authenticity of the messages exchanged between users. Explain how public and private keys would be used in your design. (5)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) A blockchain-based attendance tracking system is being developed for a small organization. Each time an employee checks in, the counter should increment, and when they check out, it should decrement. The HR manager wants to use a simple Ethereum smart contract to track the number of employees currently present in the office in real time. You are provided with the following Counter smart contract, which maintains and updates a count value using increment and decrement functions. Develop the process using this smart contract to track employee attendance. Explain how the inc() and dec() functions can be used to solve the problem of real-time employee count management on the blockchain. (5)
- A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) A developer is testing a decentralized marketplace application but encounters an error: "Invalid ABI or function signature mismatch." The developer suspects the ABI file used in the front-end is outdated. Apply your understanding of ABI structure to explain why this error occurs and how to resolve it. Describe how mismatched ABIs affect the interaction between the front-end application and the deployed contract. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) A DApp uses a contract that requires periodic upgrades. When a new version is deployed, the old contract should deactivate itself to prevent users from interacting with outdated logic. Construct a mechanism using selfdestruct()/ isActive() to retire the old contract safely while ensuring that all remaining funds are transferred to the new contract address. (5)
- A)

Course Outcome: "CO3" / Cognitive Level: "Application"

- B) A company deploys a Voting smart contract that allows users to vote for proposals. Later, they realize a bug in the vote counting logic. (5)
Build a proxy-based solution that allows the contract logic to be upgraded without losing the voting data stored on the blockchain. Explain how you would separate the storage and logic contracts.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) A developer sets up an Ethereum DApp using Remix IDE, MetaMask, and the Sepolia Test Network. The DApp compiles correctly, but transactions fail when deployed. (5)
A) **Analyze** the possible causes of deployment failure by examining the configuration between Remix, MetaMask, and the test network. Suggest what environmental dependencies could lead to such an issue.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) A university certificate verification DApp stores digital certificates on IPFS and uses Ethereum smart contracts to verify authenticity. (5)
Examine how IPFS hash values (CIDs) are used to validate certificate integrity. Differentiate this verification process from traditional centralized storage validation.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 5) A traditional escrow service relies on a middleman to hold funds until both buyer and seller fulfill their obligations. A blockchain startup replaces this with a smart contract that automatically releases funds when conditions are met. Examine how shifting from trust in a human intermediary to trust in code enhances transaction transparency and reduces fraud, but may also introduce new risks related to coding errors or vulnerabilities. (5)
A)

Course Outcome: "CO4" / Cognitive Level: "Application"

- B) A decentralized healthcare application needs to share patient test results with authorized doctors only, without exposing them publicly. (5)
Develop a mechanism using encryption or off-chain storage to ensure selective obscurity while maintaining verifiability of the data.

Course Outcome: "CO4" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4421:Ethereum Smart Contract Using Solidity

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	20
4	4	CO 4	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4421:Ethereum Smart Contract Using Solidity

#	Level	No of Questions	Marks (%)
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1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	8	40(80%)
4	Analysis	2	10(20%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)